The empirical recurrence for OEIS A283504, proved from an exact model

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Abstract

OEIS A283504 carries an empirical recurrence of order 3 contributed by Colin Barker. It is true, and it is decidable rather than empirical. The entry's sequence is given exactly by a certified growth pattern of the automaton's axis word, from which a provably satisfies a MONIC linear recurrence of order $S = 5$, derived below and not assumed. The residual of the conjectured recurrence satisfies the same one, so whether it annihilates a is settled by a finite exact computation. No transfer matrix is involved and the count is not a walk.

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1 The conjecture

OEIS A283504 is "Binary representation of the x-axis, from the left edge to the origin, of the n-th stage of growth of the two-dimensional cellular automaton defined by "Rule 641", based on the 5-celled von Neumann neighborhood.". It has offset 0 and begins

$$1, 0, 101, 1100, 11101, 111100, 1111101, 11111100, \ \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Conjectures from _Colin Barker_, Mar 10 2017: (Start) a(n) = 10*a(n-1) + a(n-2) - 10*a(n-3) for n>3.

As of the "Last modified" line on the live entry (Nov 05 2025 15:22:37, revision 11) this is still recorded as empirical, and nothing on the entry records it as proved.

2 The value is a certified growth

The entry reads an axis or a diagonal of the automaton at stage n as a string of digits and takes its numeric value. Nothing is being counted, so there is no digraph. What settles the sequence is that the CONFIGURATION is periodic in time:

$$C(n+2) \ = \ C(n) \qquad \text{as configurations of the whole plane, for every } n \geq 2.$$

This was checked directly, and one instance of it is all that is needed: the automaton is a deterministic map on configurations, so $C(n_0+2) = C(n_0)$ forces $C(n_0+k\cdot 2) = C(n_0)$ for every k, and likewise in each residue class. No locality argument and no induction over boundary windows is involved.

Getting the background right is the whole of the check. A rule taking an empty von Neumann neighbourhood to a live cell flips the ENTIRE plane, and rule 641 is read in that light: comparing two stages by padding the smaller with zeros would compare two different things.

The reading follows at once. The axis at stage n is the window $|x| \leq n$ of a configuration that is not changing, so the window widening by one cell on each side per step gives

$$w(n+2) \;=\; L_r + w(n) + R_r$$

with L_r and R_r fixed words depending only on the residue r of n modulo 2 — an identity that is now a consequence rather than an observation. Concatenation is multiplication by a power of the base B and addition, so along a class

$$a(n+2) \;=\; v(L_r)\, B^{|w(n)|+|R_r|} + a(n)\, B^{|R_r|} + v(R_r).$$

3 The bound

In the shift $T = S^2$ that is a first-order recurrence with a geometric forcing term and a constant, killed by $(T - B^{|R_r|})(T - B^{|L_r|+|R_r|})(T - 1)$. Taking the DISTINCT roots that occur across the residue classes, pulling each back to $S^2 - \lambda$, and allowing the pre-period to raise the numerator's degree gives a monic annihilator of order $S = 5$.

4 The proof

Lemma 1. *Let $A(z) = z^S - \sum_{j<S} \alpha_j z^j$ be the monic annihilator derived above and $q(z) = z^3 - \sum_i c_i z^{3-i}$ the characteristic polynomial of the conjectured recurrence. The residual $r(n) = a(n) - \sum_i c_i a(n-i)$ is a linear combination of shifts of a, so A annihilates r as well. Since $A(0) \neq 0$ the recurrence it gives runs in both directions, and S consecutive zeros of r force r to vanish at every later index.*

Proof. A annihilates a, hence every shift of a, hence any linear combination of shifts; that is r. A monic recurrence with nonzero constant term determines each term from the S before it and each from the S after it, so a run of S zeros propagates. ☐

Theorem 2.

$$a(n) \;=\; 10\, a(n-1) + a(n-2) - 10\, a(n-3)$$

for every $n > 3$.

Proof. The terms $a(0), \ldots$ were computed exactly in integer arithmetic from the model, extended by A where the model itself was not evaluated, and $r(n)$ formed for each index. Those integers vanish from $n = 3+1$ onwards, and the run exceeds $S+3$, which by Lemma 1 settles every larger index at once. The bound is exact: at $n = 3$ the recurrence fails on the entry's own published terms, so no larger range holds. ☐

5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 20 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: it is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no model involved, and holds wherever the proved range and the data overlap.

Third, the derived annihilator A was itself tested on terms it had not been given: the model was evaluated beyond the S values that determine it and A reproduced them. A bound that is too small fails this test, which is why it is run.

References

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[4] F. Diamond and J. Shurman, *A First Course in Modular Forms*, Springer GTM 228, 2005.